Summarize the functionalities of MPEG-4 and explain why it is widely used in multimedia applications.

MPEG-4 (Moving Picture Experts Group-4) is an international multimedia compression standard developed by the moving picture Experts Group. It is designed to efficiently encode audio-visual data for storage and transmission across networks

Functionalities of MPEG-4.

(1) Advanced Video Compression

- Uses efficient compression techniques
- Provides high video quality at low bit rates.
- Enables streaming over limited bandwidth networks.

(2) Audio Compression

- Supports high-quality audio coding formats.
- Offers better sound quality than older formats at similar bitrates

(3) Object-Based Coding

- Unlike earlier standards, MPEG-4 treats audio, video, text, graphics and animation as separate objects
- Allows interactive multimedia.

(4) Support for Streaming media.

- Optimized for internet streaming.
- Handles real-time transmission with error resilience

(b) Scalability.

Supports:

- Bitrate Scalability
- Spatial Scalability.
- Temporal Scalability

(c) Multimedia Integration.

- Combines video, audio, subtitles, 2D/3D graphics
- Supports interactive applications and virtual environments

(m) File format Support.

- Uses the MP4 Container format:
  - MP4.
- Stores video, audio, subtitles and metadata in a single file

Why MPEG-4 is widely used in multimedia Applications

(1) High Compression Efficiency

- Reduces file size significantly.
- Saves storage space and bandwidth.

(2) internet & streaming friendly.

- Ideal for
  - You tube
  - OTT platforms.
  - Video conferencing.
- works well over 3G/4G/5G networks

Device Compatibility.

- Support by:
  - Smartphones.
  - Smart TVs.
  - Laptops.
  - Gaming Consoles.

Flexibility & Interactivity.

- Supports multimedia-rich applications like:
  - online games.
  - E-learning
  - Virtual reality environments